Vision Aided Navigation - Exercise 6

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GitHub Repository: https://github.com/ShayMorad/3D-Computer-Vision-SLAM

(Note: Code is located on the main branch. Collaborator invites have been sent to course teachers).

Part 6.1: Relative Constraints from Local Bundle Adjustment

To transition from local sliding-window tracking to a concise global trajectory summary, the relative motion constraints and their associated uncertainties must be extracted from the local Bundle Adjustments computed in the previous exercise.

For a local window bounded by keyframes c_0 and c_k, the objective is to isolate the relative covariance $P(c_k|c_o)$. Because the local bundle adjustment mathematically anchored the first frame c_0 to the local origin using a strict PriorFactorPose3 with a minimal noise model(1×10^{-6}), the positional uncertainty of c_0 approaches zero $(\Sigma_{00} \to 0)$. Consequently, the 6×6 marginal covariance matrix extracted for the target keyframe c_k directly represents the relative spatial uncertainty between the two frames.

This extraction was executed across all dynamically selected keyframes.
The results for the very first bundle window are presented below.

```
[First Bundle: Frames 0 to 7]
Relative Pose (T_0->k):
 R: [
    0.999839, 0.0171083, -0.0053448;
    -0.0171484, 0.999825, -0.00754311;
    0.00521482, 0.00763355, 0.999957
]
t: -0.0809208  -0.104039    5.72034


Marginal Covariance Matrix (6x6):
 [[ 0.      0.      0.     -0.      0.      0.    ]
 [ 0.      0.     -0.     -0.     -0.      0.    ]
 [ 0.     -0.      0.      0.      0.      0.    ]
 [-0.     -0.      0.      0.     -0.     -0.    ]
 [ 0.     -0.      0.     -0.      0.      0.    ]
 [ 0.      0.      0.     -0.      0.      0.0001]]
```

Terminal Output Analysis: The extracted values physically validate our system parameters. The translation vector t shows a displacement of 5.72 meters along the Z-axis, perfectly aligning with our dynamic >5.0m keyframe extraction threshold (from ex5). Furthermore, the 6×6 marginal covariance matrix [Rx, Ry, Rz, Tx, Ty, Tz] accurately reflects the geometry of stereo visual odometry: angular and lateral uncertainties are incredibly small (rendering as 0. at 4 decimal places), while the largest uncertainty (0.0001) is isolated at index [5, 5]. This confirms that estimating forward depth (Z-axis translation) via stereo disparity inherently carries the highest geometric uncertainty in the system.

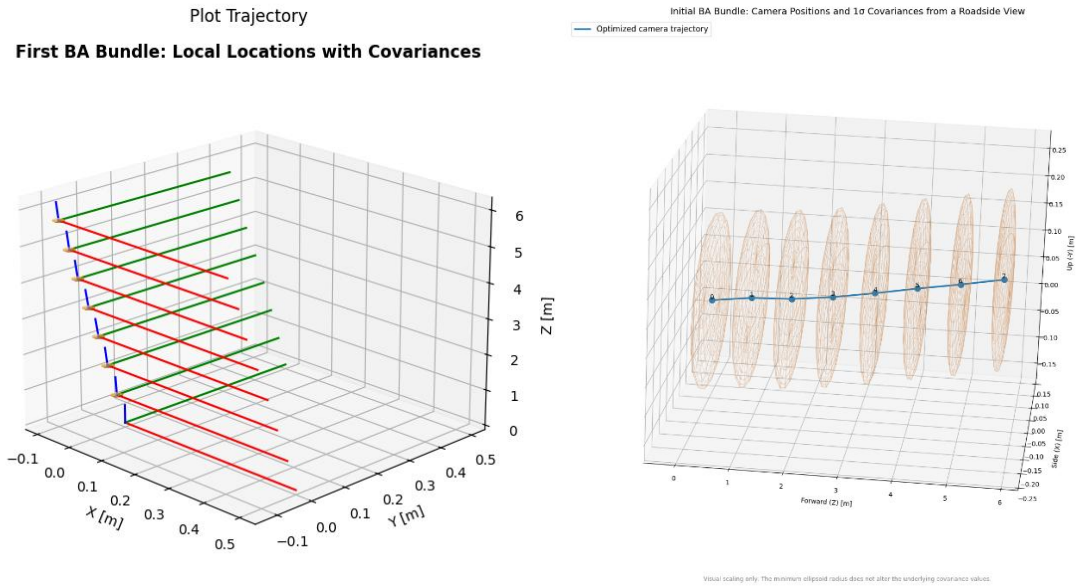


Figure 1: Visualization of the first local bundle window (Frames 0-7). The first plot displays the standard 3D trajectory and camera orientation axes, where the raw sub-millimeter uncertainty is too small to observe. The secondary "curb-side" plot isolates the translational uncertainty, applying a 150x scalar multiplier to the 1σ marginal covariance ellipsoids. This scaled perspective exposes the underlying mathematical reality: a steady accumulation of dead-reckoning drift compounding forward from the anchored origin c_0.

Code Reference: ex6.py, lines 50, 53.

Part 6.2: Global Pose Graph

With the local reprojection constraints abstracted away, a global Pose Graph was constructed. The pose graph exclusively contains BetweenFactorPose3 edges, utilizing the extracted relative transformations Δc and their corresponding marginal covariances to concisely map the 3,300-frame sequence.

- ***Question:*** *What would be a reasonable initialization for the poses? Why?*
- ***Answer:*** The most mathematically sound initialization is purely sequential kinematic chaining (dead-reckoning). By assigning the very first keyframe c_0 to the global origin $(\mathrm{I}_{4\times4})$, the global state of every subsequent keyframe c_{i+1} is computed simply by composing the previous global pose with the relative measurement constraint:

  $$c_{i+1} = c_i \cdot \Delta c_{i \to i+1}$$

  This approach is highly reasonable because it guarantees that the initial values fed into the Levenberg-Marquardt optimizer are already perfectly consistent with the odometry constraints injected into the factor graph.

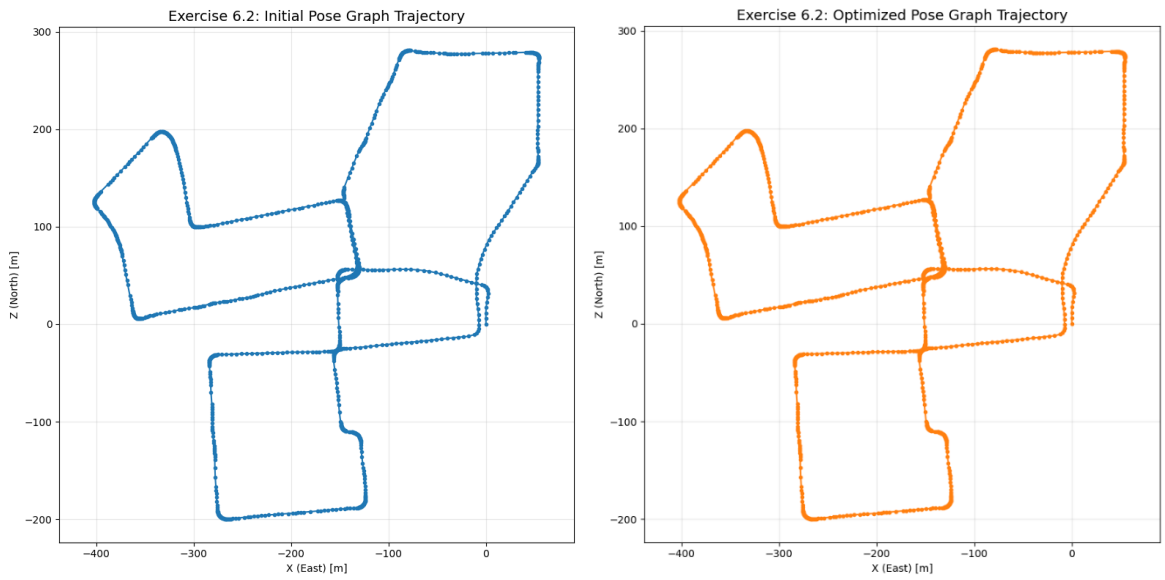


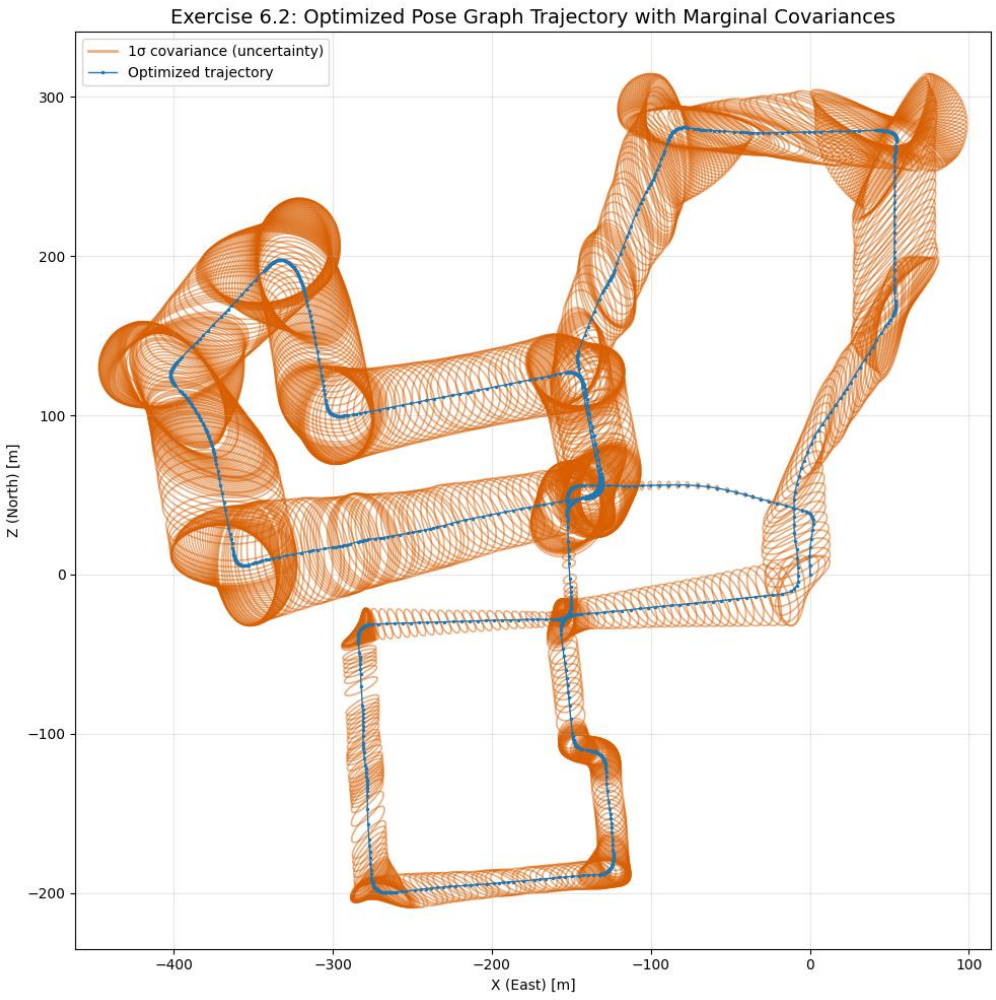
Figure 2: The initial poses supplied to the optimization, generated via sequential dead-reckoning, and the optimized poses.

Code Reference: ex6.py, lines 74, 75.

- ***Question:*** *What effect did the optimization have on the locations?*
- ***Answer:*** The global optimization had virtually zero effect on the spatial locations of the keyframes. As demonstrated in Figure 2, the optimized trajectory perfectly superimposes over the initial guess, resulting in no geometric displacement.

```
Graph Error BEFORE Optimization: 1.331597e-15
Graph Error AFTER Optimization:  7.136150e-19
```

Terminal Output Analysis: The factor graph error evaluated to 1.33e-15 before optimization and 7.13e-19 after - **both representing computational zero**. These values are entirely expected. Our global pose graph currently acts as a singular, open sequence of relative constraints. Because the system lacks global priors (e.g., absolute GPS coordinates) or loop closure factors (recognizing previously visited locations), there is zero spatial tension in the graph. Since the initial guess was generated by chaining the exact same relative measurements that act as the graph's factor constraints, the initial state inherently minimizes the cost function from step zero, leaving nothing for the optimizer to correct.



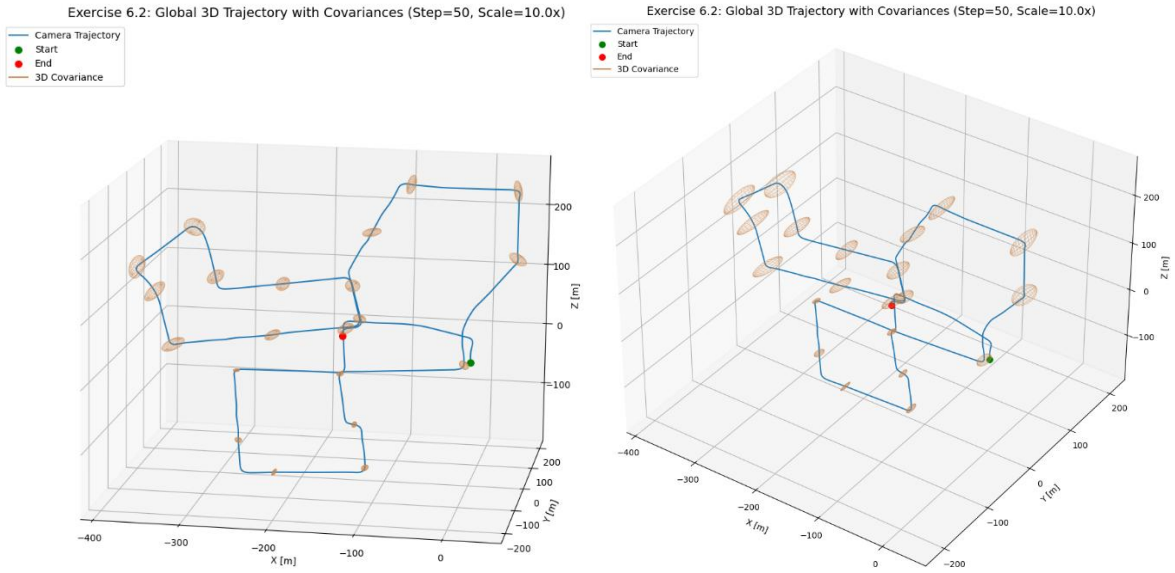


*Figure 3: Geometric visualization of global integration drift. The 2D bird's-eye view (top) plots the 1σ marginal covariances at every keyframe. To make the physical uncertainties visible on a macroscopic map spanning hundreds of meters, the **2D ellipses are artificially scaled by a factor of 30**. The 3D macro-trajectory (bottom) applies a $10\times$ scale factor and subsamples the wireframe ellipsoids every 50 keyframes. This deliberate distancing prevents visual occlusion and clearly illustrates the structural uncertainty compounding over the sequence, culminating in maximum dead-reckoning drift at the final location relative to the absolute origin.*

Code Reference: ex6.py, lines 78, 79.